Predicting Bike Rental Demand: The Impact of Weather and Feature Engineering

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1 Abstract

Bike sharing programs provide valuable data for understanding the environmental factors that influence cycling behavior. We investigate how weather and temporal factors affect daily bike rentals using *Bike Sharing Dataset* [1]. We implement linear regression using the closed-form least squares solution, achieving a baseline test MSE of 628,523. To capture non-linear relationships, we engineered polynomial, logarithmic, and interacting term features. The combined engineered model reduced test MSE to 442,662, a 30% improvement. Temperature-related features proved most informative, with cubed temperature alone reducing test MSE by 16%, which reflects human temperature preferences. Our results demonstrate that feature engineering can substantially improve linear regression performance on non-linear problems.

2 Introduction

Bike sharing programs are automatic rental systems for bicycles. These programs promote leisure time and physical activity of users without increasing amount of time spent travelling [2]. It is therefore important to understand rental behaviours, and which environmental factors influence them. Additionally, due to the automatic generation of user data from their use, bike sharing systems can be seen as a 'virtual sensor network' to sense mobility of a city and detect major events of the city [1].

The goal of our project is to model how features influence the count of bikes used daily in a Bike Sharing program, and how feature engineering can change affect performance. To examine this, we will implement linear regression with and without feature engineering, and evaluate our model's performance using Mean Squared Error.

For our data we used the *Bike Sharing Dataset* prepared by Hadi Fanaee-T and Joao Gama. Capital Bikeshare system in Washington D.C., USA provides a two-year log for the years 2011 and 2012, documenting user activity (`http://capitalbikeshare.com/system-data`). The Bike Sharing Dataset from Laboratory of Artificial Intelligence and Decision Support (LIAAD), University of Porto combines this Capital Bikeshare data with weather and season information from `http://www.freemeteo.com` [1].

3 Data Preprocessing and Exploration

We first inspected our dataset and found 731 samples with 15 features, not including the target `cnt`. We also investigated the data type of each of the features.

We then searched the dataset for any missing values, which could cause failures or errors in our model. We used `pandas.DataFrame.dropna()` which removes `NaN`, `None`, or `NaT` values, but found 0 missing values. Next, we dropped irrelevant features including `instant` (the ID of the data point), `dteday` (the full date), and `atemp` (in favour of `temp` that does not already factor in humidity and windspeed). We dropped `workingday`, because this

information is already contained in the features `weekday` and `holiday`. We dropped `season`, because this information is contained in the feature `mnth`. Finally, we dropped `casual` and `registered`, which sum to our feature of interest `count` and would cause data leakage. This left us with 8 features (not including `cnt`).

For numerical features, we observed that the data had already been normalized by division of their maximum value. Therefore, we decided not to do additional standardization. For categorical features with more than 2 categories, we one-hot encoded the categorical features `mnth` (12), `weekday` (7), and `weathersit` (3) to avoid a false ordering of the data. This left us with 27 total features (not including `cnt`).

We visualized our data using `matplotlib` and `seaborn` in order to inform our modelling decisions. We observed that the numerical features have different distributions: `temp` appears more normal, `hum` appears left-skewed, and `windspeed` appears right-skewed (Figure 3). We observed that rental distributions vary significantly by season and weather, suggesting that these are important predictive features. Rental distributions did not vary significantly by weekday or workingday. The distribution of rentals shifted right from the first year of data to the second (Figure 6), indicating the system was more popular in the second year. We observed that `temp` has a positive correlation with `cnt`. We also observed `hum` and `windspeed` both had poor linear correlations with `cnt` (Figure 4).

4 Methods

4.1 Linear Regression Implementation

We implemented linear regression, trained using the closed-form least squares solution with `NumPy`. Our model learns weight such that:

$$\hat{y} = Xw$$

$$\begin{aligned} X &: \text{feature matrix} \\ w &: \text{vector of learned coefficients} \\ \hat{y} &: \text{predictions} \end{aligned}$$

During training (fit), the input features are first reshaped if needed, and a column of ones is optionally added to account for the bias (intercept) term.

$$y = \begin{bmatrix} x & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_0 \end{bmatrix}$$

We solve least squares using `np.linalg.lstsq(a,b,rcond=None)` to compute the weight vector that best fits the training data by minimizing the squared prediction error.

$$w = (X^T X)^{-1} X^T y$$

We made predictions using our learned weights. The learned weight vector contains one weight per feature (and one additional weight for the bias, if enabled). During prediction (predict), the same feature formatting is applied to the input data, and predictions are computed as a simple matrix multiplication between the input matrix and the learned weights.

$$\hat{y} = Xw$$

We prepared our dataset for supervised learning by separating input features (X) from the target variable (y), then randomly splitting the data into training (80%) and test sets (20%), wth random seed 2 for reproducibility. The linear regression model is trained only on the training data, and the test set is held out for final evaluation.

We created a function to evaulate the Mean Squared Error (MSE) of our model to observe its performance.

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

4.2 Feature Engineering

We engineered features to recognize potential non-linear relationships in the data, that we might expect to see. Extreme cold and heat can impair the human body. Therefore we used `temp`^{2} and `temp`^{3} to capture the impact of human temperature preference. Similarly, as humidity increases, we might expect to see a more pronounced impact, therefore we use `hum`^{2}. Likewise, we also tested `windspeed`^{2}.

We expect the combined effect of humidity and temperature to negatively affect bike riding conditions, so we used `temp*hum` to try and account for this interaction. Log transformations of temperature (`log(temp)`) and humidity (`log(hum)`) were also tested to capture potential diminishing returns between those features and `cnt`.

Finally, in our last model edit we combined all the engineered features that caused a decrease in MSE of the testing and training sets.

Table 1: Summary of engineered features.

Engineered feature	**Definition**
Squared humidity	`hum`^{2}
Squared temperature	`temp`^{2}
Squared windspeed	`windspeed`^{2}
Temp–humidity interaction	`temp*hum`
Log-temperature	`log(temp)`
Log-humidity	`log(hum)`

5 Results

5.1 Baseline Model

Predicted counts correlate with actual counts from the test data. Points close to the perfect prediction line indicate somewhat accurate predictions. The distribution of prediction errors appears normal and centered around 0, which indicates an unbiased model. However, there is a large variance and room for model improvement. The MSE of the training set was 650464.4744 and the MSE of the test set was 628522.9338.

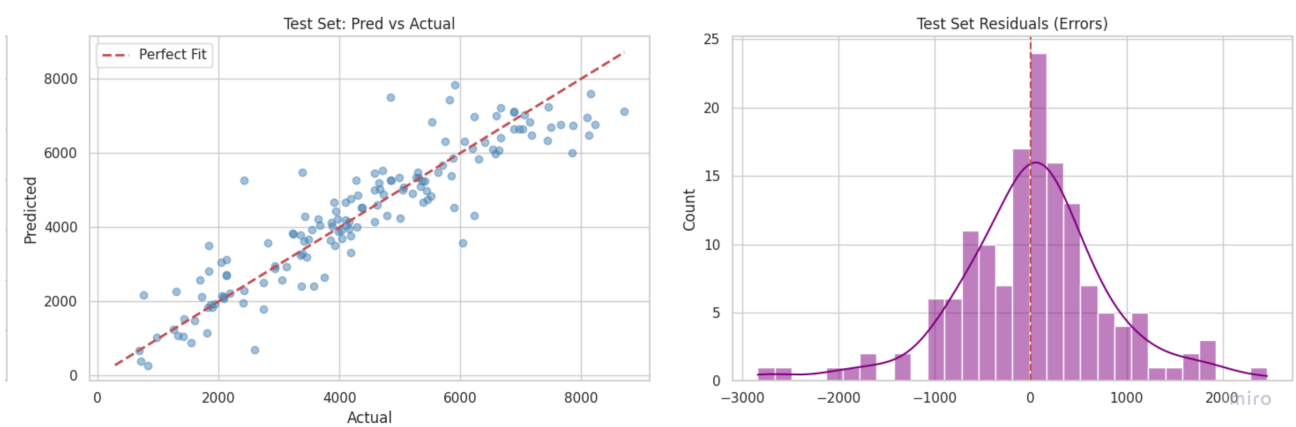

Figure 1: Evaluation of baseline linear regression model

5.2 Engineered Model

All engineered features except for `temp*hum` had a decrease in MSE (Table 2). The engineered feature with the largest impact was `temp`^{3} with a reduced test MSE of 16%. The final model was a combination of the the engineered features had a decrease in train MSE of 30%.

Table 2: Performance comparison of linear regression with feature engineering

Model variant	Feature-engineering	Train MSE	Test MSE	Δ Train	Δ Test
Baseline	None	650464.4744	628522.9338	–	–
+ Squared humidity	$\texttt{hum}^2$	639924.6198	624222.5247	+10539.8546	+4300.4091
+ Squared temperature	$\texttt{temp}^2$	565000.1877	562281.8910	+85464.2867	+66241.0427
+ Cubed temperature	$\texttt{temp}^3$	544713.1601	525278.5186	+105751.3143	+103244.4151
+ Squared windspeed	$\texttt{windspeed}^2$	647203.3744	628491.7090	+3261.1000	+31.2247
+ Temp–humidity interaction	`temp*hum`	626159.2318	659514.3879	+24305.2426	-30991.4541
+ Log-temperature	`log(temp)`	579623.2250	586712.3763	+70841.2494	+41810.5575
+ Log-humidity	`log(hum)`	640233.8160	623522.1134	+10230.6584	+5000.8204
Combination		489225.4054	442661.9927	+161239.0689	+185860.9411

Figure 2: Evaluation of engineered linear regression model

6 Discussion and Conclusion

1. Effect of preprocessing choices on model performance and stability:

- **One-Hot Encoding significantly improved model expressiveness:** Converting categorical features (`mnth`, `weathersit`, `weekday`) into binary variables was critical. Without this, the linear regression model would have treated these as continuous ordinal values (e.g., treating December as "12 times" January), which would have degraded performance by imposing incorrect linear relationships on nominal data.

- **Removal of redundant features improved stability:** By dropping features like `atemp` (highly correlated with `temp`) and `season` (redundant with `mnth`), we reduced multicollinearity. This improves the numerical stability of the closed-form solution $(X^T X)^{-1} X^T y$ and makes the learned coefficients more interpretable.

- **Leakage Prevention ensured validity:** Dropping `casual` and `registered` was a necessary choice to ensure the model predicted `cnt` based on environmental factors rather than simply summing up the target's components, ensuring the reported test MSE reflects true generalization capability.

2. Features that appeared the most informative:

- **Temperature (`temp`) was the most dominant predictor:** In our feature engineering experiments, the $\texttt{temp}^2$ transformation yielded the largest reduction in Test MSE ($+66,241$ improvement), far outperforming changes to humidity or windspeed. This suggests a strong non-linear relationship where user preference peaks at a specific temperature range.

- **Interaction terms were less effective:** While individual features like temperature were informative, complex interaction terms like $\texttt{temp} \times \texttt{hum}$ led to overfitting (increasing Train performance but degrading Test performance by $-30,991$), suggesting that simpler features were more robust for this dataset.

- **Year** (`yr`) showed significance (Figure 6). The difference between the first and second year reflect growing popularity of the Bikeshare system.

3. Effect of non-linear feature engineering on training error:

- Adding non-linear terms reduced the training MSE from 650,464.47 (baseline) to 489,225.41 (combined feature engineered model), a decrease of 161,239.07 ($\approx 24.8\%$). Individually, higher-order temperature terms were most impactful (e.g., $\texttt{temp}^3$ reduced train MSE by 105,751.31), indicating important non-linear effects.

4. Limitations of linear regression for this dataset:

- **Limited ability to model complex patterns:** Even after adding some engineered features, linear regression still mostly combines features in an additive way. This makes it hard to capture more complex relationships, such as weather affecting bike rentals differently across seasons. As a result, some non-linear patterns in the data may still be underfit.

- **Sensitivity to extreme values:** Because the model is trained using mean squared error, days with very high or very low rental counts can strongly influence the learned weights. This can lead to poorer performance on more typical days and larger errors at the extremes.

- **Correlated features:** Adding polynomial, logarithmic, and interaction features can introduce strong correlations between predictors. This makes the learned coefficients harder to interpret and can lead to unstable weights, even if the overall training error improves.

- **Mismatch with count data:** Bike rentals are non-negative integer counts, but linear regression can produce negative or non-integer predictions. This means the model does not naturally respect the structure of the target variable.

5. Possible next steps:

- Further standardization could be beneficial so that the different variances of features did not cause them to have disproportionate effects.

- More sophisticated models such as decision trees or gradient boosting could capture non-linear relationships without manual feature engineering

- Future work could explore regularization techniques (Ridge or Lasso regression) to address potential overfitting from the expanded feature set

- Cross-validation could improve performance estimates over a single train-test split

7 Statements

- **Statement of contributions**: All team members contributed equally.
- **Statement of resource usage**: We used Prism (previously crixet) for LaTeXformatting

8 Appendix A

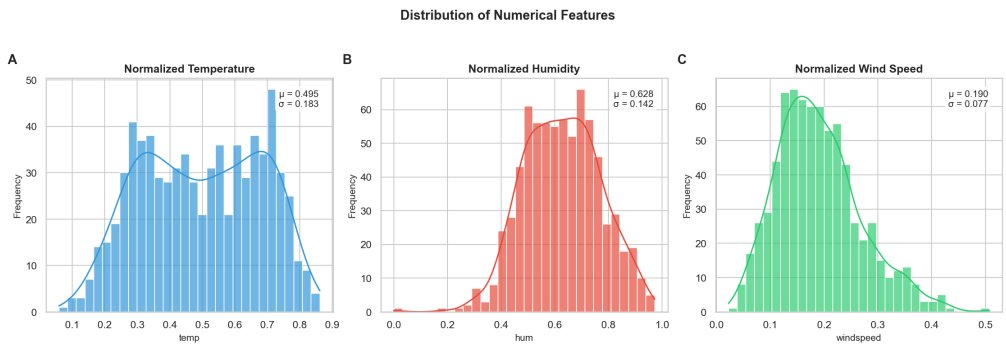

Figure 3: Histogram distributions of numerical features temp, hum, and windspeed with KDE.

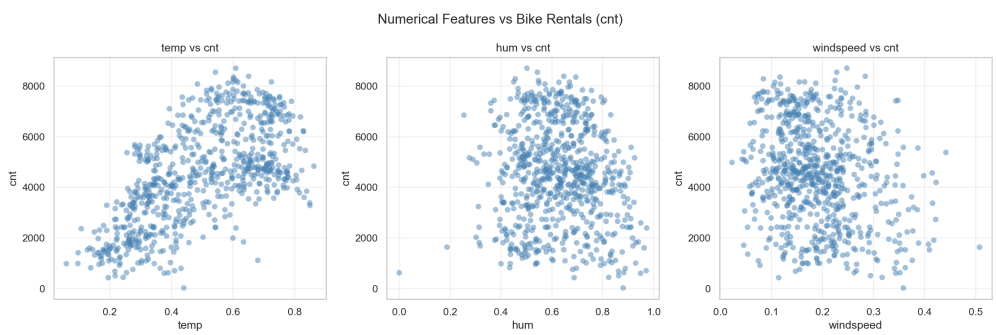

Figure 4: Numerical features plotted against daily bike count.

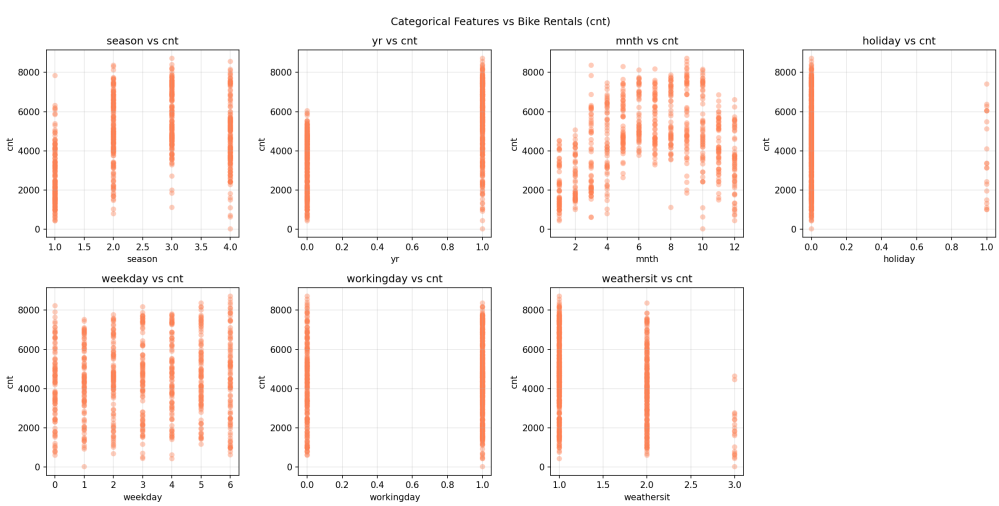

Figure 5: Categorical features plotted against daily bike count.

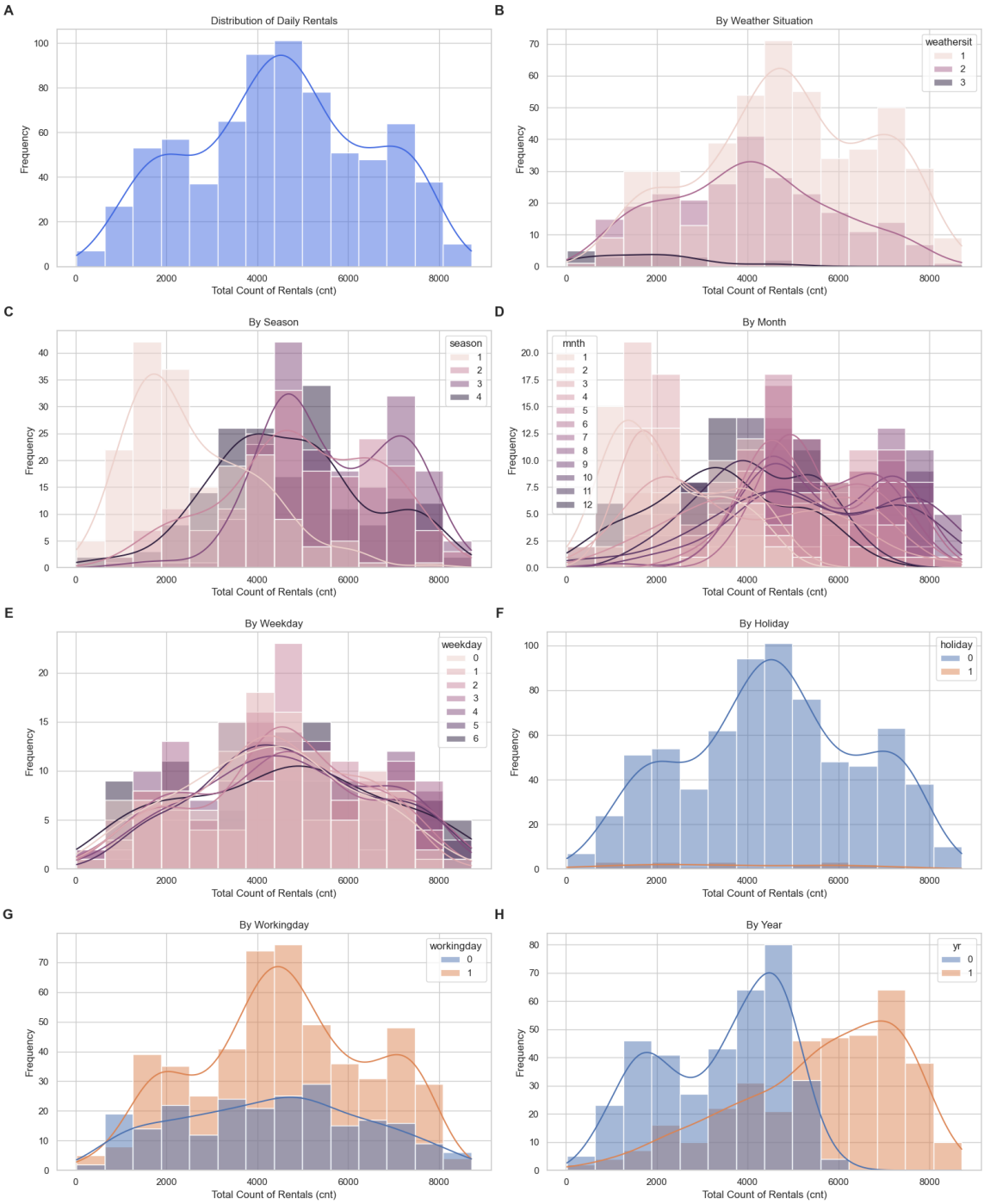

Figure 6: Histogram of our target feature cnt with KDE, coloured according to the different categories for each categorical feature.

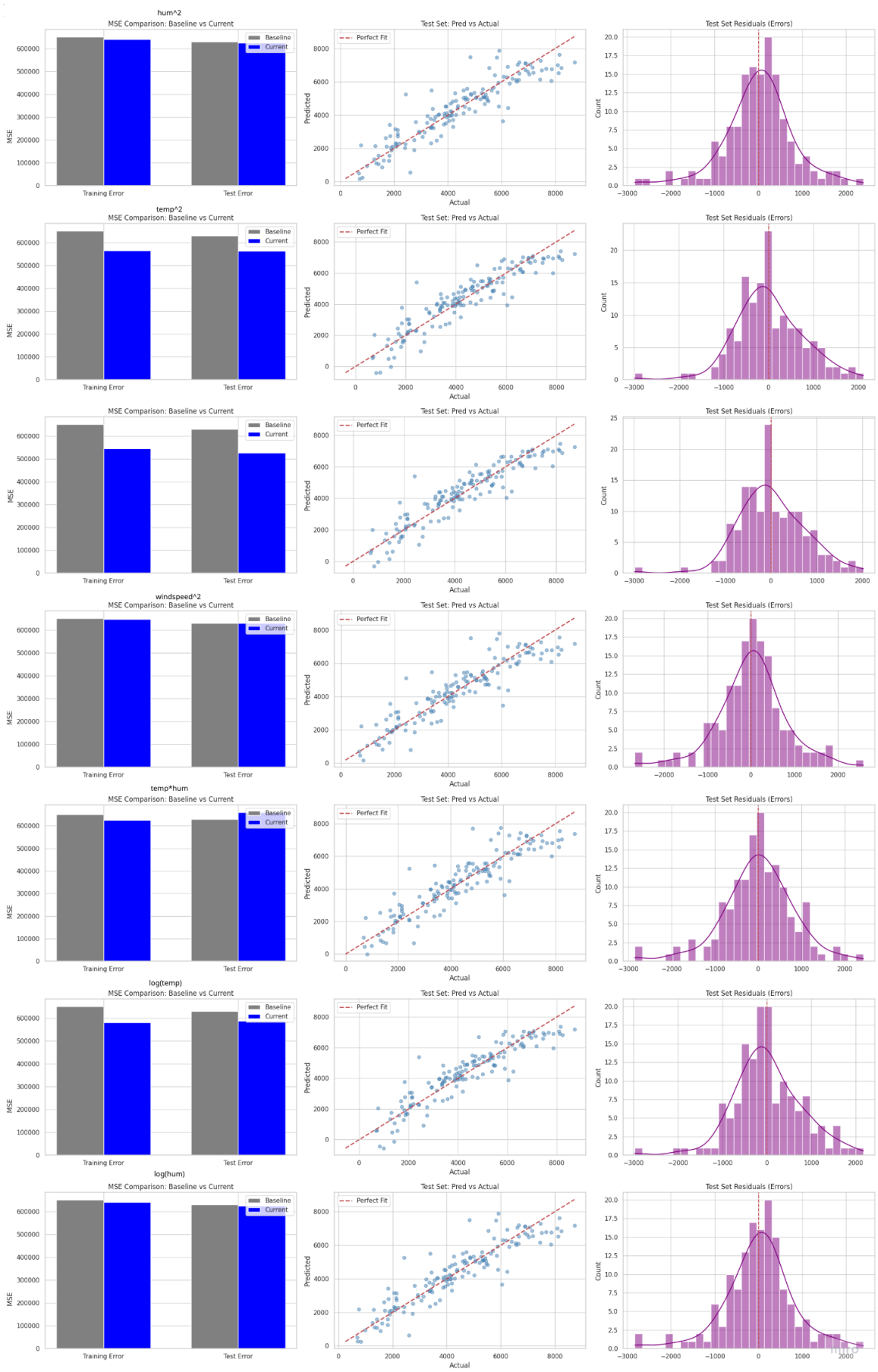

Figure 7: Evaluation of the impact of each engineered feature on model performance

References

[1] Hadi Fanaee-T and Joao Gama. Event labeling combining ensemble detectors and background knowledge. *Progress in Artificial Intelligence*, pages 1–15, 2013.

[2] Lin Lin, Sun-Young Kim, Daniel Yonto, and Zihao Chen. Pedaling towards health: exploring the impact of bike sharing systems on physical activity in 15-minute neighborhoods. *Transportation Research Interdisciplinary Perspectives*, 34:101781, November 2025.